

can trigger a faster heart rate, sweating, or fear, even if there is currently nothing scary or threatening in your surroundings. Such storage and recall of emotional memory are especially dependent on function of the **amygdala**, an almond-shaped brain structure near the base of the cerebrum.

Often, generating emotion and experiencing emotion require interactions between different regions of the brain. For example, both laughing and crying involve the limbic system interacting with sensory areas of the forebrain. Similarly, structures in the forebrain attach emotional “feelings” to survival-related functions controlled by the brainstem, including aggression, feeding, and sexuality.

To study the function of the human amygdala, researchers sometimes present adult subjects with an image followed by an unpleasant experience, such as a mild electrical shock. After several trials, study participants experience *autonomic arousal*—as measured by increased heart rate or sweating—if they see the image again. Subjects with brain damage confined to the amygdala can recall the image because their explicit memory is intact. However, they do not exhibit autonomic arousal, indicating that damage to the amygdala has resulted in a reduced capacity for emotional memory.

Functional Imaging of the Brain

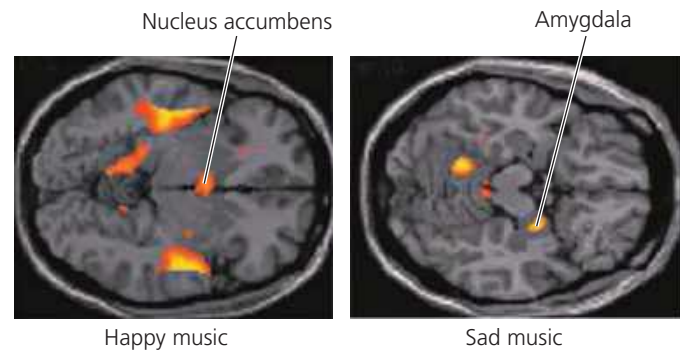
In recent years, scientists have begun studying the amygdala and other brain structures with functional imaging techniques. By scanning the brain while the subject performs a particular function, such as forming a mental image of a person’s face, researchers are able to match particular functions with activity in specific brain areas.

Several approaches are available for functional imaging. The first widely used technique was positron-emission tomography (PET), in which injection of radioactive glucose enables a display of metabolic activity. Today, the most commonly used approach is functional magnetic resonance imaging (fMRI). In fMRI, a subject lies with his or her head in the center of a large, doughnut-shaped magnet. Brain activity is detected by an increase in the flow of oxygen-rich blood into a particular region.

In one experiment using fMRI, researchers mapped brain activity while subjects listened to music that they described as sad or happy (**Figure 49.15**). The findings were striking: Different regions of the brain were associated with the experience of each of these contrasting emotions. Subjects who heard sad music had increased activity in the amygdala. In contrast, listening to happy music led to increased activity in the *nucleus accumbens*, a brain structure important for the perception of pleasure.

Functional imaging has an ever-increasing number of applications. Hospitals use fMRI to, for example, monitor recovery from stroke, map abnormalities in migraine headaches, and increase the effectiveness of brain surgery.

▼ **Figure 49.15 Functional imaging in the working brain.** These images resulted from using fMRI to reveal brain activity associated with music that listeners described as happy or sad. (Each view shows activity in a single plane of the brain, as seen from above.)



VISUAL SKILLS The two images reveal activity in different horizontal planes through the brain. How can you tell this from the two photographs? What can you conclude about the location of the nucleus accumbens and the amygdala?

CONCEPT CHECK 49.2

1. When you wave your right hand, what part of your brain initiates the action?
2. People who are inebriated have difficulty touching their nose with their eyes closed. Which brain region does this observation indicate is one of those impaired by alcohol?
3. **WHAT IF?** Suppose you examine two groups of individuals with CNS damage. In one group, the damage has resulted in a coma (a prolonged state of unconsciousness). In the other group, it has caused paralysis (a loss of skeletal muscle function throughout the body). Relative to the position of the midbrain and pons, where is the likely site of damage in each group? Explain.

For suggested answers, see Appendix A.

CONCEPT 49.3

The cerebral cortex controls voluntary movement and cognitive functions

We turn now to the cerebrum, the part of the brain essential for language, cognition, memory, consciousness, and awareness of our surroundings. As shown in Figure 49.11, the cerebrum is the largest structure in the human brain. Like the brain overall, it exhibits regional specialization. For the most part, cognitive functions reside in the cortex, the outer layer of the cerebrum. Within this cortex, *sensory areas* receive and process sensory information, *association areas* integrate the information, and *motor areas* transmit instructions to other parts of the body.

In discussing the location of particular functions in the cerebral cortex, neurobiologists often use four regions, or *lobes*, as physical landmarks. Each lobe—frontal, temporal, occipital, and parietal—is named for a nearby bone of the skull, and each is the focus of specific brain activities (**Figure 49.16**).